usb_joystick_routed

Design Report

Rev 1 | 2026-10-18 | jlcpcb-tier1

kicad-tools 0.23.0

Board Summary

Property	Value
Layers	4 copper (F.Cu, In1.Cu, In2.Cu, B.Cu)
Footprints	38 (37 SMD, 1 THT, 0 other)
Nets	27
Traces	1080 segments
Vias	105
Board Size	80.0 x 60.0 mm

Stackup

Layer	Type	Thickness (mm)	Material
F.Cu	copper	0.035	--
dielectric 1	prepreg	0.2104	7628
In1.Cu	copper	0.0152	--
dielectric 2	core	1.065	FR4
In2.Cu	copper	0.0152	--
dielectric 3	prepreg	0.2104	7628
B.Cu	copper	0.035	--

Design Overview

Theory of Operation

USB joystick controller — ATmega32U4

USB bus powered, 8MHz crystal, AVR ISP programming

GPIO and ADC map is fixed to Microchip TQFP-44 pinout

Communication Interfaces

Protocol	Signals
SPI	ISP_MISO, ISP_MOSI, ISP_SCK
USB	USB_D+, USB_D-, VBUS

Assembly Notes

1 fine-pitch component

- **Fine-pitch components**: 1 (U1)

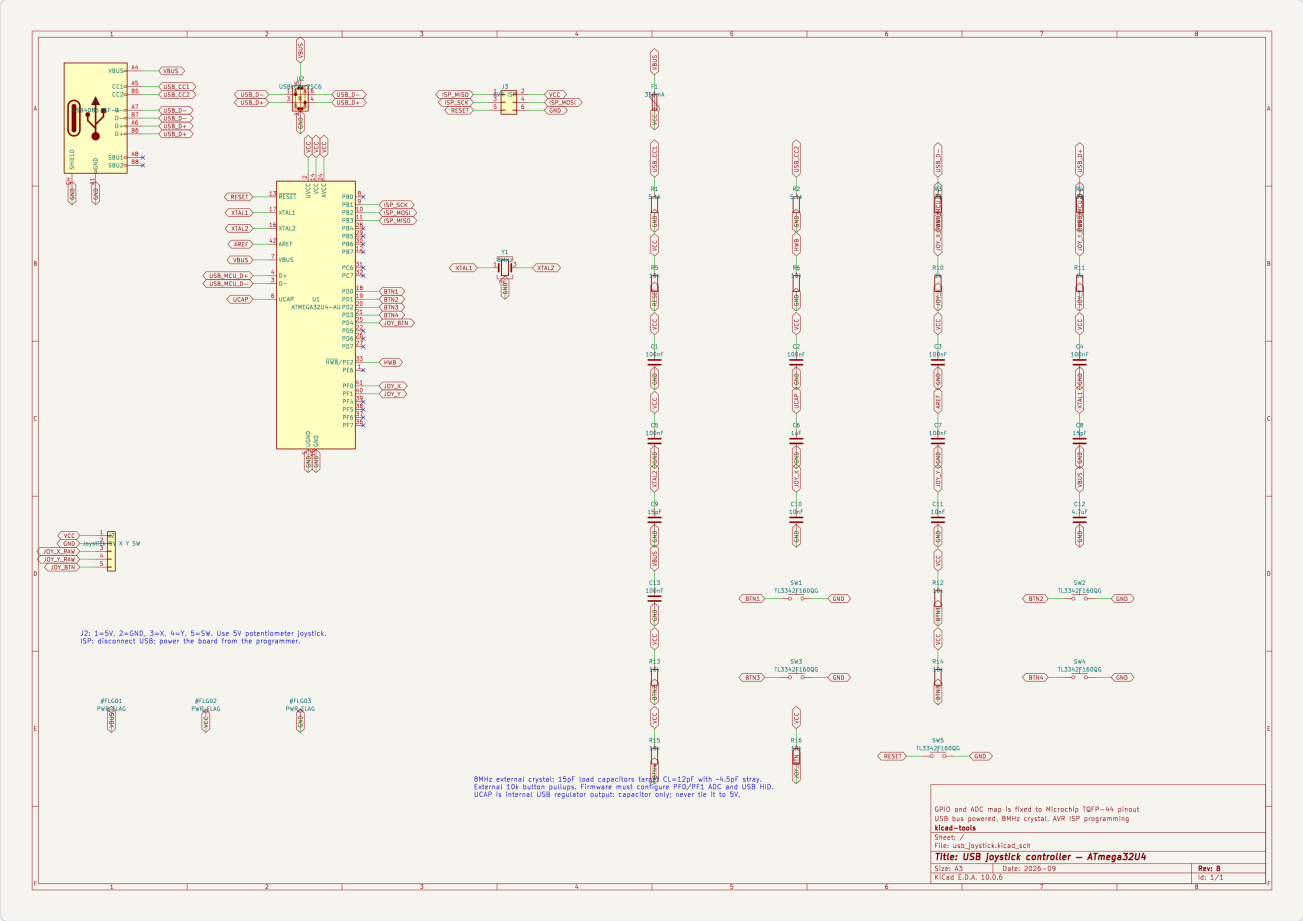
ERC Status

Metric	Count
Errors	0
Warnings	0

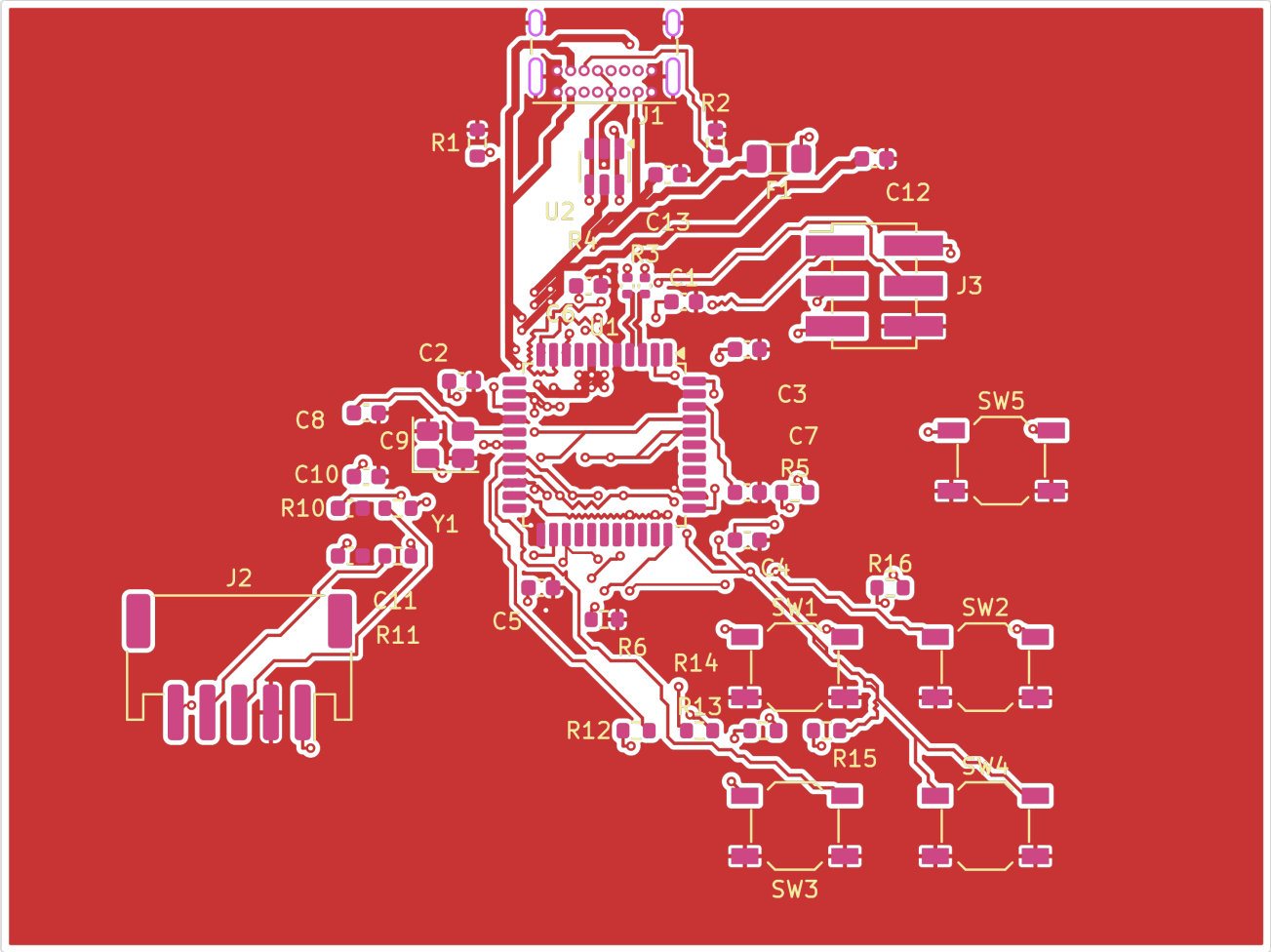
Status: PASS

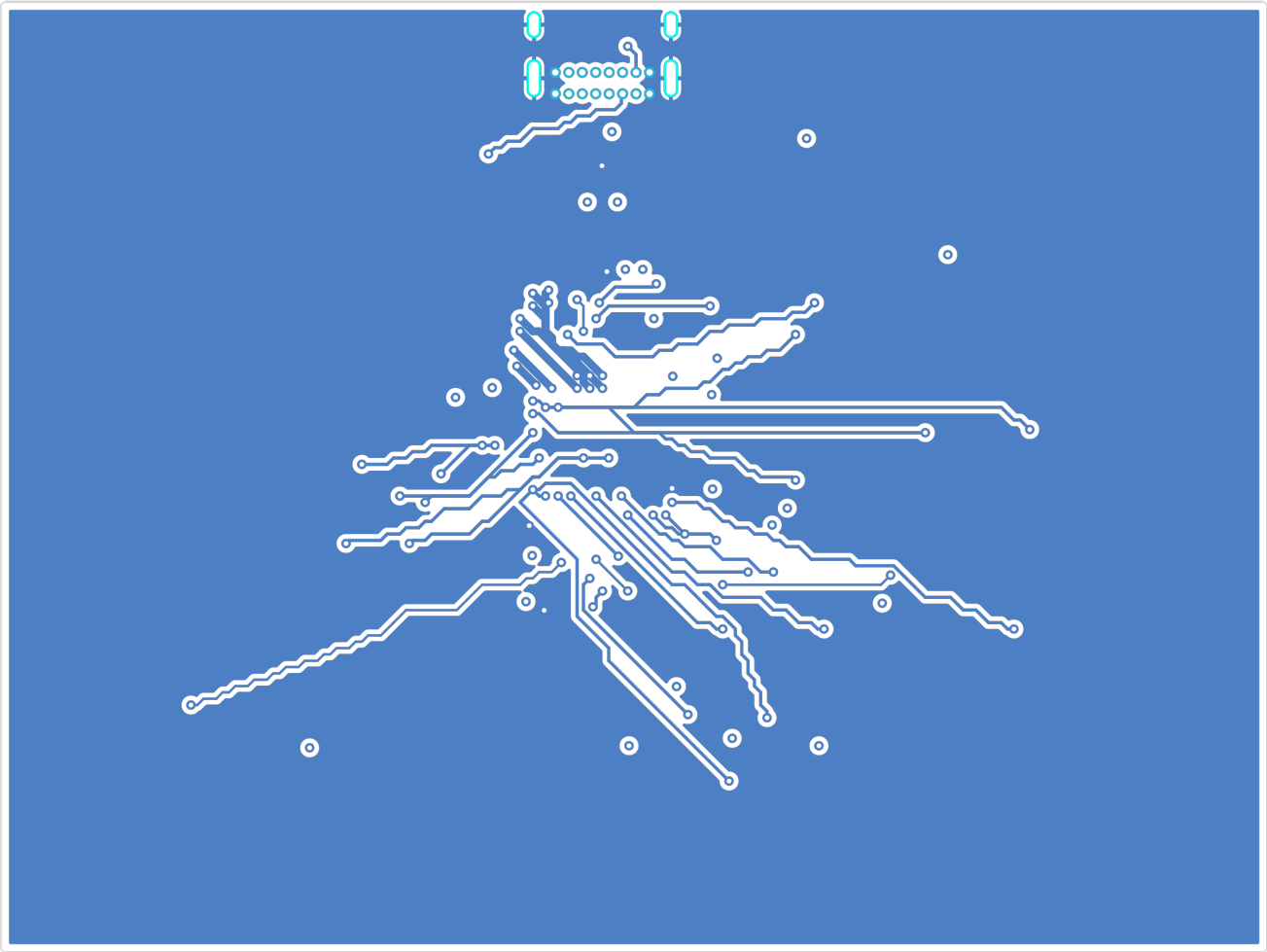
Schematic Overview

Schematic: usb_joystick

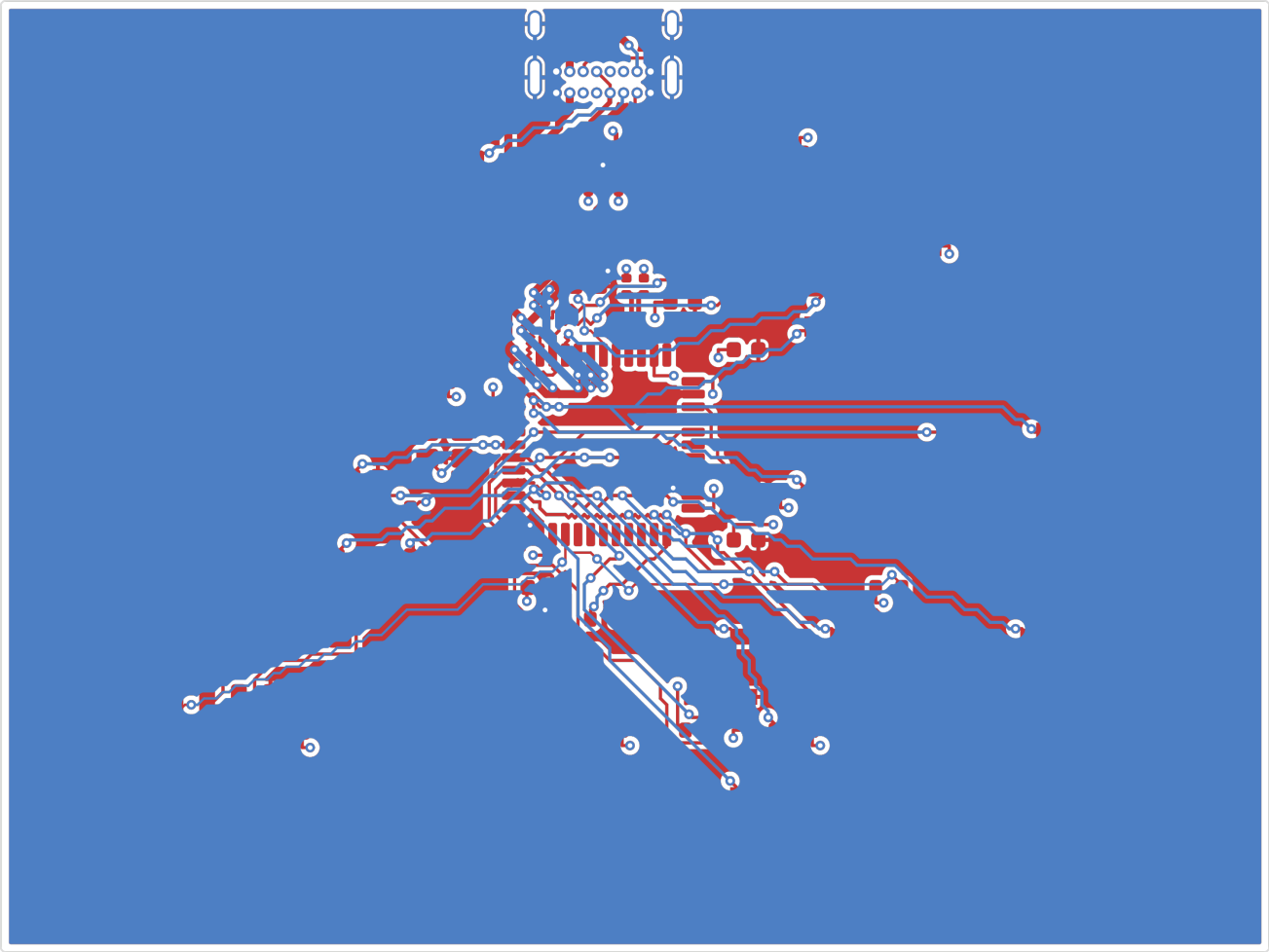


PCB Layout

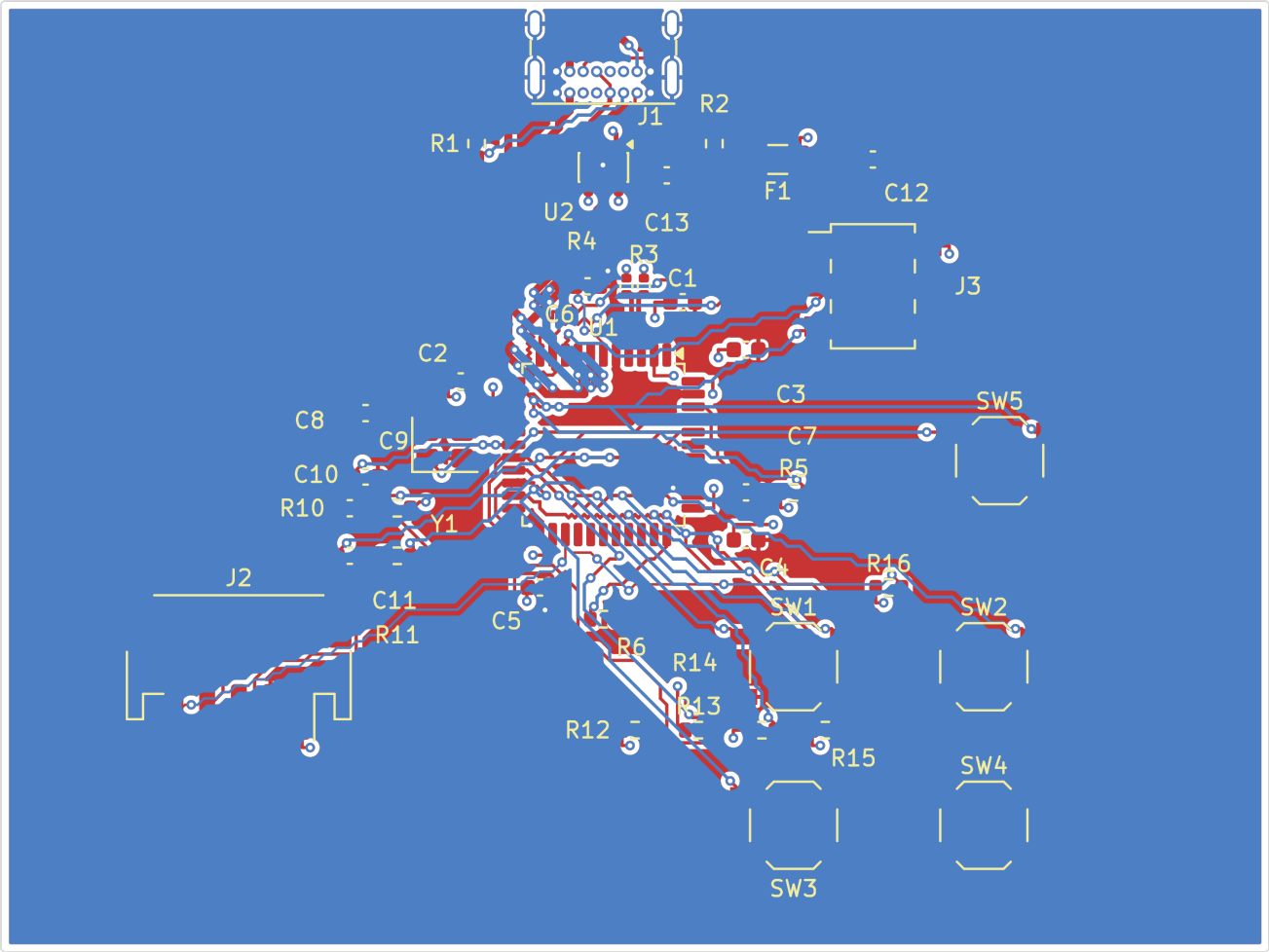

R1
J1
R2
U2
C12
C13
R3
J3
C1
C2
C3
SW5
C8
C9
C7
C10
R5
R10
Y1
J2
C4
R16
SW1
SW2
C11
C5
R11
R6
R14
R13
R12
R15
SW4
SW3



Copper

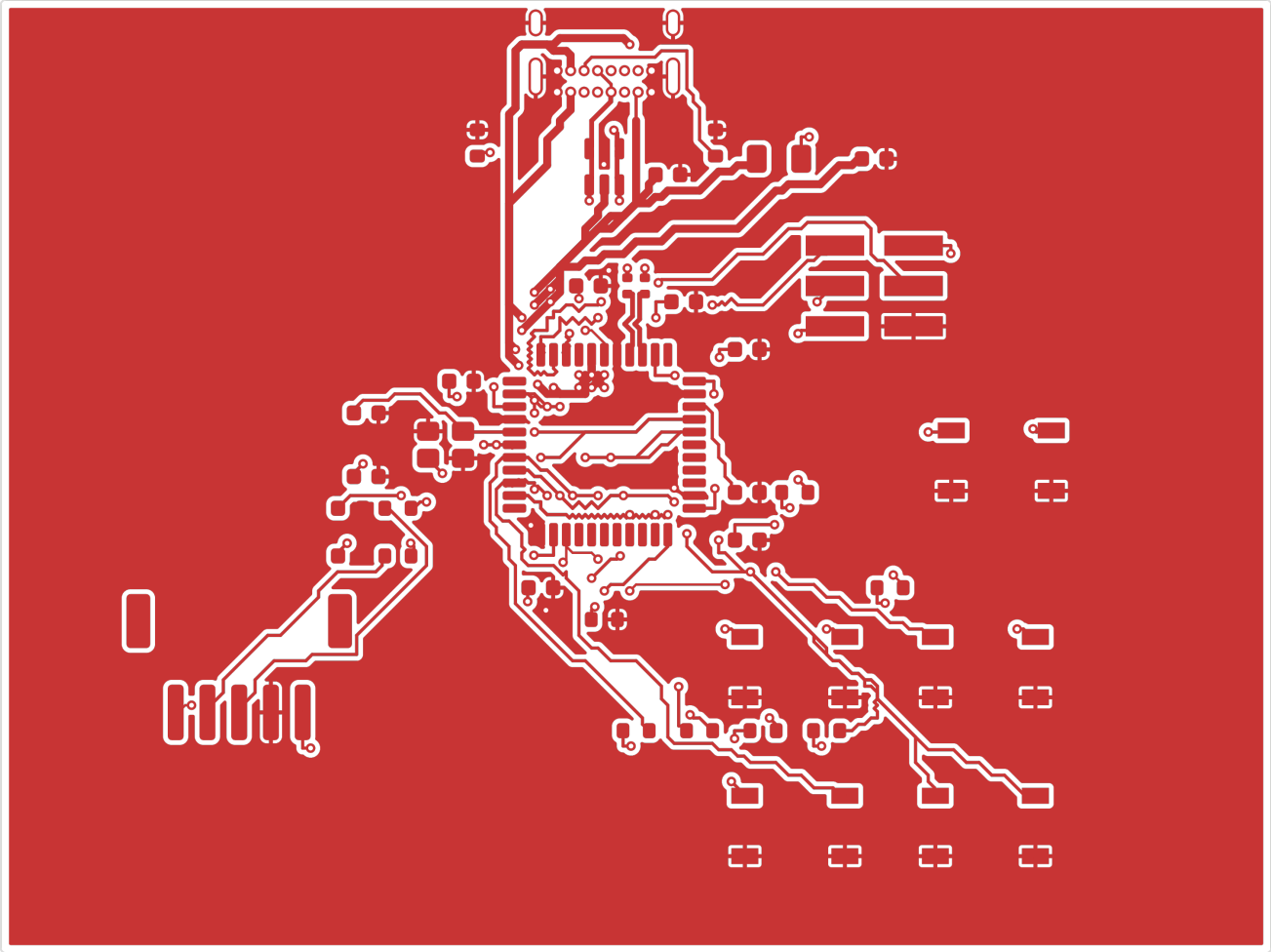


Assembly

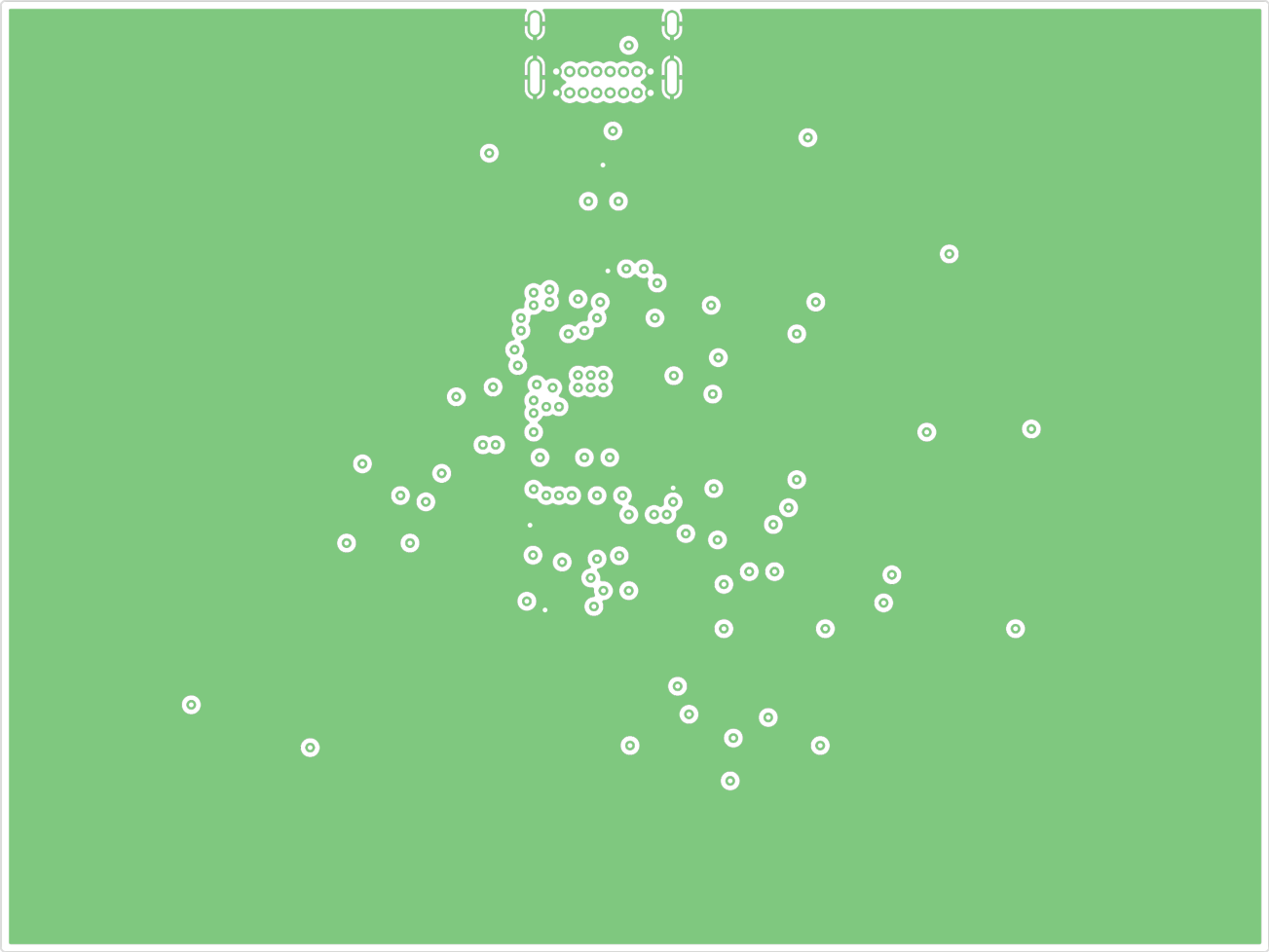

J1
R1
R2
U2
F1
C12
C13
R4
R3
C1
J3
C6
C2
C3
SW5
C8
C9
C7
C10
R5
R10
Y1
J2
C11
R16
SW1
SW2
C5
R11
R6
R14
R13
R12
R15
SW4
SW3

Copper Layers

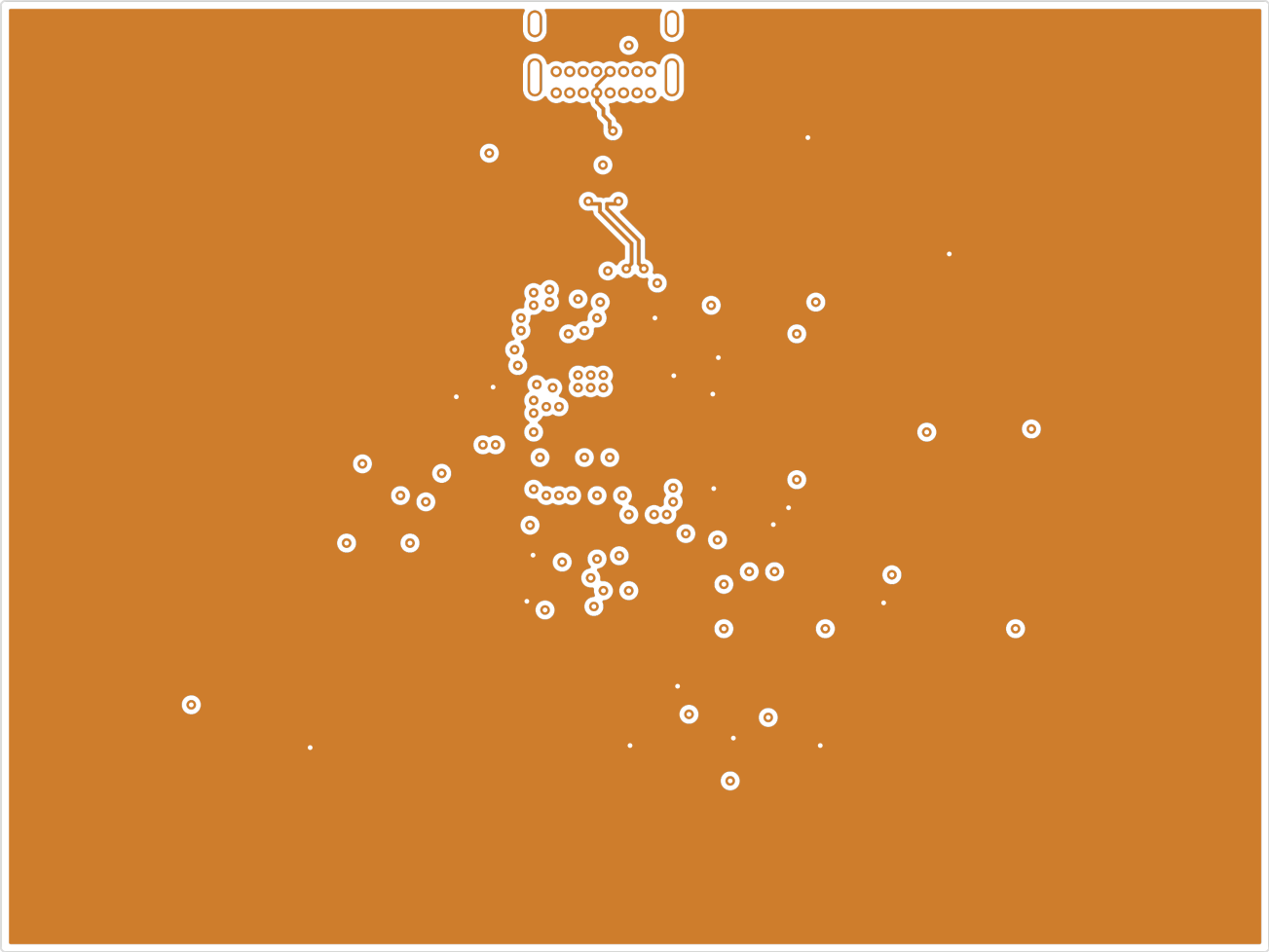
F.Cu



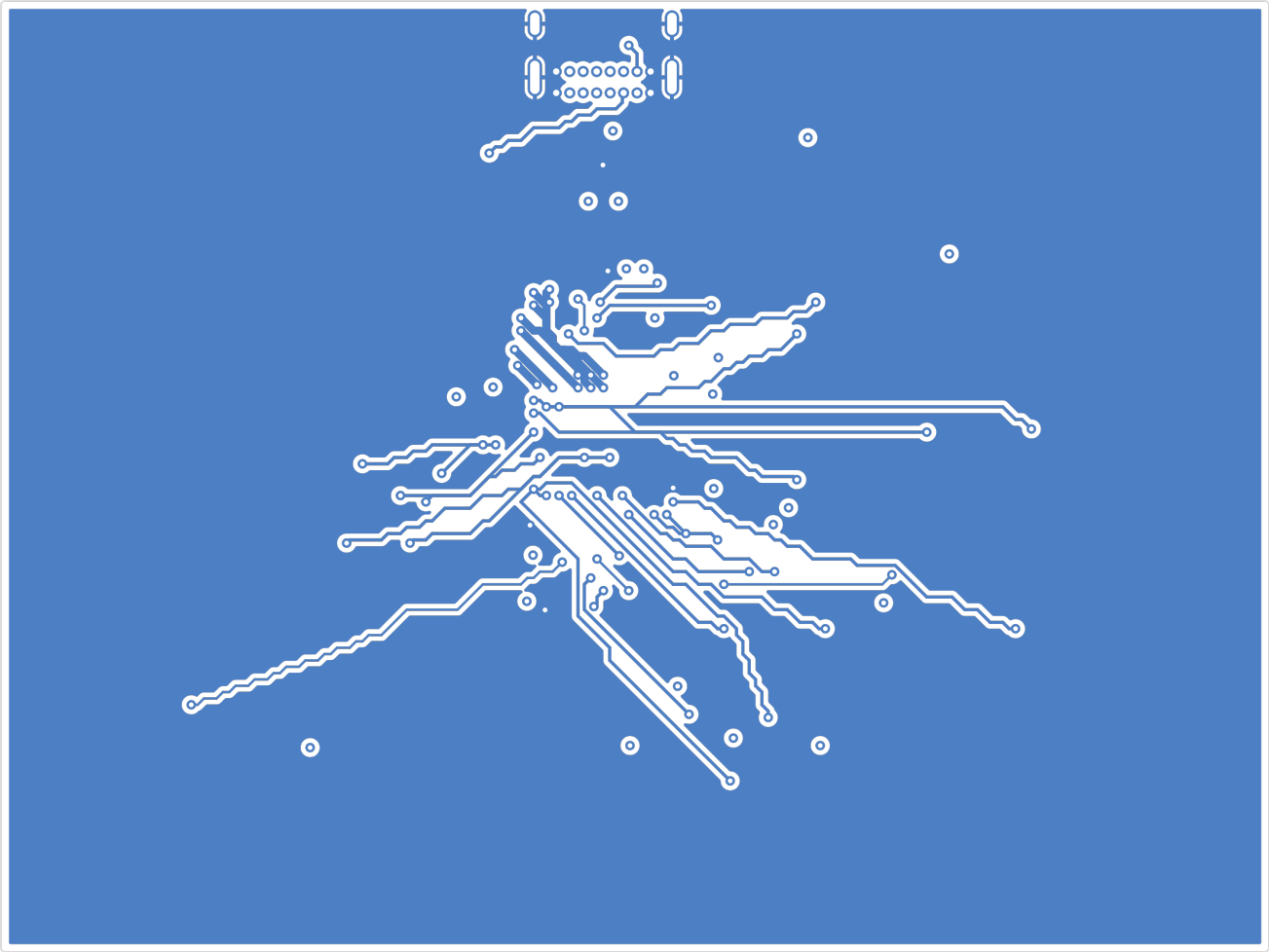
In1.Cu



In2.Cu



B.Cu



usb_joystick_routed

Bill of Materials

Value	Package	Qty	References	MPN	LCSC
100nF	C_0603_1608 Metric	7	C1, C2, C3, C4, C5, C7, C13	GRM188R71C 104KA01D	C45000
10nF	C_0603_1608 Metric	2	C10, C11	GRM188R71H 103KA01D	C77053
15pF	C_0603_1608 Metric	2	C8, C9	GRM1885C1H 150JA01D	C71651
1uF	C_0603_1608 Metric	1	C6	GRM188R71A 105KA61D	C97888
4.7uF	C_0603_1608 Metric	1	C12	GRM188R61A 475KE15D	C71633
350mA	Fuse_1206_32 16Metric	1	F1	1206L035/16Y R	C126817
AVR ISP	Samtec_TSM-103-01-T-DV	1	J3	TSM-103-01-T-DV-P-TR	C21285200
Joystick 5V X Y SW	JST_PH_S5B-PH-SM4-TB_1x05-1MP_P2.00mm_Horizontal	1	J2	S5B-PH-SM4-TB(LF)(SN)	C265104
USB4085-GF-A	USB_C_Receptacle_GCT_USB4085	1	J1	USB4085-GF-A	C7095263
10k	R_0603_1608 Metric	7	R5, R6, R12, R13, R14, R15, R16	RC0603FR-07 10KL	C98220
1k	R_0603_1608 Metric	2	R10, R11	RC0603FR-07 1KL	C22548
22	R_0402_1005 Metric	2	R3, R4	RC0402FR-07 22RL	C114765
5.1k	R_0603_1608 Metric	2	R1, R2	RC0603FR-07 5K1L	C105580
TL3342F160Q G	SW_SPST_TL3342	5	SW1, SW2, SW3, SW4, SW5	TL3342F160Q G	C2886898

Value	Package	Qty	References	MPN	LCSC
ATMEGA32U4-AU	TQFP-44_10x10mm_P0.8mm	1	U1	ATMEGA32U4-AU	C44854
USBLC6-2SC6	SOT-23-6	1	U2	USBLC6-2SC6	C7519
8MHz	Crystal_SMD_3225-4Pin_3.2x2.5mm	1	Y1	ECS-80-12-33-JGN-TR	C2442665

DRC Status (`--mfr jlcpcb-tier1` rules)

Metric	Count
Errors	0
Errors: kct (`--mfr jlcpcb-tier1` rules)	0
Errors: kicad-cli (geometric)	0
Warnings	21
Blocking	0

Status: PASS

Violations by Type

Violation Type	Count
hole_to_hole_clearance	14
thermal_asymmetry	6
diffpair_routing_continuity	1
diffpair_uncoupled_length	1

Manufacturing Readiness

Verdict: WARNING

Action Items

- **[OPTIONAL]** Refill zones to verify 2 zone-connected nets, then re-run kct audit
- **[OPTIONAL]** Review 21 DRC warnings

Routing Status

Metric	Value
Signal Net Completion	100.0% (25/25)
Overall Completion	100.0%
Complete Nets	27 / 27
Zone-Connected Nets	2
Incomplete Nets	0
Unconnected Pads	0

Zone-Connected Nets

- GND
- VCC

Cost Estimate

Metric	Per Board (estimated)
PCB Fabrication	~2.96 USD
Components (estimated)	~2.11 USD
Assembly (estimated)	~2.16 USD
Total (estimated)	**~7.23 USD**
Batch Quantity	5
Batch Total (estimated)	~36.14 USD